☑ CTL Provisional DECISION SHEET (PDSH)
☐ CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	**Subclause(s)**	**Tracking No.**
IEC 62841-1: 2014 IEC 62841-2-3: 2020	Clause 21.18.1	2270
Category		
TOOL		
Subject	**Keywords**	**Developed by**
Power Switches for hand-held tools	Power switch, hand-held tool, flexible shaft	ETF 1

Question

A given power tool is formed by an angle grinder motor unit which is connected by means of a flexible shaft to a sanding head. The length of flexible shaft is approximately 1 m.

The sanding head is intended for polishing, sanding and wire brushing operations. It can be equipped with discs having a rated capacity > 100mm, which perform rotary motion only.

As per the instruction manual, the sanding head has to be operated using both hands, while the angle grinder motor unit is carried by the user body by means of a strap or belt. The power switch which controls the on/off function of the angle grinder motor unit and sanding head is located on the angle grinder motor unit and can be locked in the “on” position.

Is the above construction in compliance with clause 21.18.1 of IEC 62841-1:2014 and IEC 62841-2-3:2020?

Decision

The above-described construction does not meet the requirements of the standards (Clause 21.18.1).

Explanatory notes

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DSH

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Clause 21.18 of IEC 60745-1:2006 states that tools other than those provided with a flexible shaft shall be fitted with a mains switch which can be switched on and off by the user without releasing his grasp on the Tool. This exception related to flexible shafts has been deleted in IEC 62841-1:2014.

Clause 21.18.1 of IEC 62841-1 only specifies that the power switch must be operable without the user releasing any handles or grasping surfaces required for safe handling during use.
According to clause 19.4 of IEC 62841-2-3:2020, the tool must be controlled using two hands because it has a rated capacity > 100 mm.

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Given the flexible shaft between the angle grinder motor unit and the sanding head, it is evident that both hands must be placed on the sanding head for proper handling (and for fulfilling clause 19.4 of IEC 62841-2-3:2020).

Due to the position of the power switch, the user needs to release one hand from the grasping surfaces to switch the tool on or off.